Penlight

STUDY GUIDE

Cardiac Conduction & Dysrhythmias

Penlight sample lecture · 12 min lecture · October 17, 2026

Lite mode (offline) Sample content

5	28	47	17
SECTIONS	KEY TERMS	FLASHCARDS	PRACTICE QS

Overview

This lecture (about 12 min, 1,906 words) covers Conduction System, Electrocardiogram, Sinus Rhythms and Ventricular Dysrhythmias. Normal sinus rhythm has a rate of sixty to one hundred, a regular rhythm, a P wave before every QRS, a normal PR interval, and a narrow QRS. Remember that the ventricular rhythms with no pulse require defibrillation, the unstable rhythms with a pulse require synchronized cardioversion, and the stable rhythms are treated with medications.

What you should be able to do

- Explain the main ideas in “Conduction System”.
- Explain the main ideas in “Electrocardiogram”.
- Explain the main ideas in “Sinus Rhythms”.
- Explain the main ideas in “Ventricular Dysrhythmias”.
- Explain the main ideas in “Heart Blocks”.
- Define asystole in your own words.
- Define atrial flutter in your own words.
- Define automaticity in your own words.
- Recall the priority and safety points the instructor emphasized.

Must-know points

- A patient can have an organized rhythm on the monitor and still have no pulse, so always assess the patient first. 3:08
- The key idea is to treat the cause, not the number on the monitor. 5:33
- The ventricles quiver and do not pump, so there is no cardiac output and no pulse. 8:15
- Pulseless electrical activity, or PEA, refers to organized electrical activity on the monitor with no palpable pulse. 9:57
- Remember that the ventricular rhythms with no pulse require defibrillation, the unstable rhythms with a pulse require synchronized cardioversion, and the stable rhythms are treated with medications. 11:56

Lecture notes

1 Conduction System 0:00

- The impulse normally begins in the sinoatrial node, also called the SA node. 0:48
- When the SA node is working and the rhythm originates there, we call it a sinus rhythm. 1:15
- From the SA node, the impulse travels across both atria, causing atrial depolarization and atrial contraction. 1:20
- The AV node's intrinsic rate is forty to sixty beats per minute, and it serves as the backup pacemaker if the SA node fails. 1:47
- The Purkinje fibers carry the impulse rapidly through the ventricular muscle so that both ventricles contract together. 2:09

ventricle | SA node | P wave | AV node | pacemaker | depolarization

2 Electrocardiogram 2:52

> A patient can have an organized rhythm on the monitor and still have no pulse, so always assess the patient first. 3:08

- The PR interval is measured from the start of the P wave to the start of the QRS complex, and it represents the time for the impulse to travel from the atria through the AV node. 3:19
- The QRS complex represents ventricular depolarization, and a normal QRS is less than zero point twelve seconds. 3:35

> The QT interval is measured from the start of the QRS to the end of the T wave, and a prolonged QT interval increases the risk of a dangerous rhythm called torsades de pointes. 3:51

pulse | ventricle | P wave | AV node | PR interval | depolarization

3 Sinus Rhythms 4:45

- Normal sinus rhythm has a rate of sixty to one hundred, a regular rhythm, a P wave before every QRS, a normal PR interval, and a narrow QRS. 4:45
- Sinus bradycardia is a sinus rhythm with a rate below sixty. 4:56
- Sinus tachycardia is a sinus rhythm with a rate above one hundred. 5:23

> The key idea is to treat the cause, not the number on the monitor. 5:33

- In atrial fibrillation the atria quiver chaotically instead of contracting, so there are no distinct P waves, and the ventricular response is irregularly irregular. 5:44

ventricle | P wave | PR interval | asystole | sinus rhythm | atrial flutter

4 Ventricular Dysrhythmias 7:05

- Ventricular tachycardia is a rapid rhythm of three or more PVCs in a row, with a wide QRS and a rate above one hundred. 7:37
- If the patient has no pulse, ventricular tachycardia is treated exactly like ventricular fibrillation. 8:10

> The ventricles quiver and do not pump, so there is no cardiac output and no pulse. 8:15

- We use cardioversion for rhythms in which the patient has a pulse. 8:47

> Pulseless electrical activity, or PEA, refers to organized electrical activity on the monitor with no palpable pulse. 9:57

pulse · ventricle · SA node · P wave · ventricular tachycardia · asystole

5 Heart Blocks 10:13

- Heart blocks are problems with conduction through the AV node. 10:19
- First-degree AV block is defined as a prolonged PR interval greater than zero point twenty seconds, with every impulse still conducted. 10:24
- Second-degree AV block type two, also called Mobitz two, has a constant PR interval with some P waves that are not followed by a QRS. 10:46
- Third-degree AV block, also called complete heart block, means that no impulses pass from the atria to the ventricles. 11:02

> Remember that the ventricular rhythms with no pulse require defibrillation, the unstable rhythms with a pulse require synchronized cardioversion, and the stable rhythms are treated with medications. 11:56

pulse · ventricle · P wave · AV node · pacemaker · PR interval

Glossary

Asystole	The absence of any electrical activity.
Atrial fibrillation	In atrial fibrillation the atria quiver chaotically instead of contracting, so there are no distinct P waves, and the ventricular response is irregularly irregular.
Atrial flutter	A rapid, regular atrial rhythm that produces a sawtooth pattern on the ECG.
Automaticity	The ability of cardiac cells to generate an electrical impulse on their own without being told to by the brain.

AV dissociation	The atria and ventricles beat independently.
AV node	Heart blocks are problems with conduction through the AV node.
Cardioversion	A shock delivered in time with the QRS complex, which is why it is called synchronized.
Defibrillation	The delivery of an unsynchronized electrical shock to depolarize the entire heart so that the SA node can take over again.
Depolarization	The electrical change that triggers the muscle cell to contract.
Dysrhythmia	Atrial fibrillation is the most common sustained dysrhythmia.
ECG	A graphic recording of the heart's electrical activity.
First-degree AV block	A prolonged PR interval greater than zero point twenty seconds, with every impulse still conducted.
P wave	The PR interval is measured from the start of the P wave to the start of the QRS complex, and it represents the time for the impulse to travel from the atria through the AV node.
Pacemaker	A device that delivers an electrical impulse to stimulate the heart when the natural conduction system fails.
PR interval	The PR interval is measured from the start of the P wave to the start of the QRS complex, and it represents the time for the impulse to travel from the atria through the AV node.
Pulse	Remember that the ventricular rhythms with no pulse require defibrillation, the unstable rhythms with a pulse require synchronized cardioversion, and the stable rhythms are treated with medications.
Purkinje fibers	The Purkinje fibers carry the impulse rapidly through the ventricular muscle so that both ventricles contract together.
QRS complex	The QRS complex represents ventricular depolarization, and a normal QRS is less than zero point twelve seconds.
Repolarization	The return of the cell to its resting electrical state after depolarization.
SA node	From the SA node, the impulse travels across both atria, causing atrial depolarization and atrial contraction.

Second-degree AV block	Second-degree AV block type two, also called Mobitz two, has a constant PR interval with some P waves that are not followed by a QRS.
Sinus bradycardia	A sinus rhythm with a rate below sixty.
Sinus rhythm	Sinus tachycardia is a sinus rhythm with a rate above one hundred.
Sinus tachycardia	A sinus rhythm with a rate above one hundred.
T wave	The QT interval is measured from the start of the QRS to the end of the T wave, and a prolonged QT interval increases the risk of a dangerous rhythm called torsades de pointes.
Ventricle	The ventricles quiver and do not pump, so there is no cardiac output and no pulse.
Ventricular fibrillation	If the patient has no pulse, ventricular tachycardia is treated exactly like ventricular fibrillation.
Ventricular tachycardia	A rapid rhythm of three or more PVCs in a row, with a wide QRS and a rate above one hundred.

Quick review checklist

- [ ] A patient can have an organized rhythm on the monitor and still have no pulse, so always assess the patient first.
- [ ] The key idea is to treat the cause, not the number on the monitor.
- [ ] The ventricles quiver and do not pump, so there is no cardiac output and no pulse.
- [ ] Pulseless electrical activity, or PEA, refers to organized electrical activity on the monitor with no palpable pulse.
- [ ] Remember that the ventricular rhythms with no pulse require defibrillation, the unstable rhythms with a pulse require synchronized cardioversion…
- [ ] The impulse normally begins in the sinoatrial node, also called the SA node.
- [ ] Normal sinus rhythm has a rate of sixty to one hundred, a regular rhythm, a P wave before every QRS, a normal PR interval…
- [ ] Ventricular tachycardia is a rapid rhythm of three or more PVCs in a row, with a wide QRS and a rate above one hundred.

Flashcard preview

Asystole
The absence of any electrical activity.

Atrial flutter
A rapid, regular atrial rhythm that produces a sawtooth pattern on the ECG.

Automaticity
The ability of cardiac cells to generate an electrical impulse on their own without being told to by the brain.

AV dissociation
The atria and ventricles beat independently.

Cardioversion
A shock delivered in time with the QRS complex, which is why it is called synchronized.

Defibrillation
The delivery of an unsynchronized electrical shock to depolarize the entire heart so that the SA node can take over again.

Depolarization
The electrical change that triggers the muscle cell to contract.

ECG
A graphic recording of the heart's electrical activity.

First-degree AV block
A prolonged PR interval greater than zero point twenty seconds, with every impulse still conducted.

Pacemaker
A device that delivers an electrical impulse to stimulate the heart when the natural conduction system fails.

Self-test

1. Which term is described here? “The absence of any electrical activity.”

A. Asystole
B. Second-degree AV block
C. Cardioversion
D. Sinus rhythm

2. Which term is described here? “A rapid rhythm of three or more PVCs in a row, with a wide QRS and a rate above one hundred.”

A. AV node
B. Ventricular tachycardia
C. Pacemaker
D. Advanced cardiac life support

3. Which term is described here? “The atria and ventricles beat independently.”

A. AV dissociation
B. Third-degree AV block
C. P wave
D. Cardioversion

4. Which term is described here? “The return of the cell to its resting electrical state after depolarization.”

A. QRS complex
B. Second-degree AV block
C. Repolarization
D. Dysrhythmia

5. Which term is described here? “A prolonged PR interval greater than zero point twenty seconds, with every impulse still conducted.”

A. Sinus tachycardia
B. First-degree AV block
C. Asystole
D. Chest pain

6. Which term is described here? “A rapid, regular atrial rhythm that produces a sawtooth pattern on the ECG.”

A. Purkinje fibers
B. P wave
C. Atrial flutter
D. Chest pain

7. Which term is described here? “A graphic recording of the heart's electrical activity.”

A. AV node
B. First-degree AV block
C. ECG
D. Depolarization

8. According to the lecture, what is Cardioversion?

A. A rapid, regular atrial rhythm that produces a sawtooth pattern on the ECG.
B. The absence of any electrical activity.
C. A shock delivered in time with the QRS complex, which is why it is called synchronized.
D. A prolonged PR interval greater than zero point twenty seconds, with every impulse still conducted.

Answer key

1. A — The lecture defines Asystole: “The absence of any electrical activity.” (≈9:36)

2. B — The lecture defines Ventricular tachycardia: “A rapid rhythm of three or more PVCs in a row, with a wide QRS and a rate above one hundred.” (≈7:37)

3. A — The lecture defines AV dissociation: “The atria and ventricles beat independently.” (≈11:07)

4. C — The lecture defines Repolarization: “The return of the cell to its resting electrical state after depolarization.” (≈4:02)

5. B — The lecture defines First-degree AV block: “A prolonged PR interval greater than zero point twenty seconds, with every impulse still conducted.” (≈10:24)

6. C — The lecture defines Atrial flutter: “A rapid, regular atrial rhythm that produces a sawtooth pattern on the ECG.” (≈6:16)

7. C — The lecture defines ECG: “A graphic recording of the heart's electrical activity.” (≈2:57)

8. C — “Cardioversion” — A shock delivered in time with the QRS complex, which is why it is called synchronized. (≈8:42)

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